Project Proposal

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A capstone proposal on the development of a DFM workbench for FreeCAD
Engineering Research Preparation

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October 29, 2025

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1 Introduction

In modern product development, the practice of Design for Manufacturability (DFM) is a critical discipline that bridges the gap between digital design and physical production. It is an engineering methodology focused on designing products for ease and economy of manufacturing, directly impacting cost, quality, and time-to-market. Computer-Aided Design (CAD) software serves as the primary environment where these principles are applied, and the automation of DFM checks within these platforms has become a key driver of efficiency.

While mature commercial CAD suites like SOLIDWORKS and Fusion 360 have long integrated automated DFM analysis tools, a significant capabilities gap exists within the open-source ecosystem. FreeCAD, a leading open-source parametric modeler with a growing global user base of hobbyists, students, and small enterprises, currently lacks a dedicated, integrated workbench for DFM analysis. This absence forces users to rely on manual, error-prone inspection or expensive third-party services, creating a barrier to producing high-quality, manufacturable designs and undermining one of the key advantages of using a streamlined, digital workflow.

This research proposal outlines the development of a novel, integrated DFM Workbench for FreeCAD to address this critical gap. This capstone project's primary goal is to create a modular, extensible, and user-friendly tool that provides designers with automated, interactive, and actionable feedback. The initial scope will focus on the well-defined rules for injection moulding, with a core architectural design that explicitly allows for future expansion to other manufacturing processes. The project aims to deliver not just a functional tool, but a robust platform that empowers the open-source community with capabilities previously confined to proprietary software.

This proposal is structured to provide a comprehensive overview of the research plan. Chapter 2 presents a literature review of the core DFM principles for injection moulding, establishing the theoretical foundation for the automated checks. Chapter 3 synthesises these findings to formulate a precise research question. Chapter 4 details the Design Science Research methodology that will guide the project through a five-phase development cycle. Finally, Chapter 5 concludes by summarising the project's anticipated contributions to both engineering practice and the body of design knowledge for open-source tools.

2 | Literature Review

2.1 Overview

Design for manufacturing (DFM) is a crucial process in any products development. It is defined as the engineering practice of designing products for ease and economy of manufacturing (Boothroyd, Dewhurst, & Knight, 2002). As computer-aided design (CAD) tools have become increasingly powerful, there has been a corresponding rise in automated tools to assist engineers in applying DFM principles to their designs (Gupta, Das, Regli, & Nau, 1995). The concept for this project emerged from foundational discussions on the official FreeCAD developer communication channels in early June 2025. During these exchanges, members of the FreeCAD Development Team (The FreeCAD Team, 2025) identified a need for automated design rule checks, framing it as a high-value DFM capability that was absent in the open-source ecosystem. The objective of the literature review is to conduct an analysis of the current landscape of DFM, in terms of existing automated tool and foundation standards that have long existed. Specifically, this review will focus on the manufacturing process of injection moulding, a well-established method of mass producing plastic parts with well defined standards and processes (Kauffer, 2011). The findings will be synthesised to justify and inform the development of an extensible DFM workbench for FreeCAD, an open-source, community-driven, parametric 3D CAD program (The FreeCAD Team, 2025).

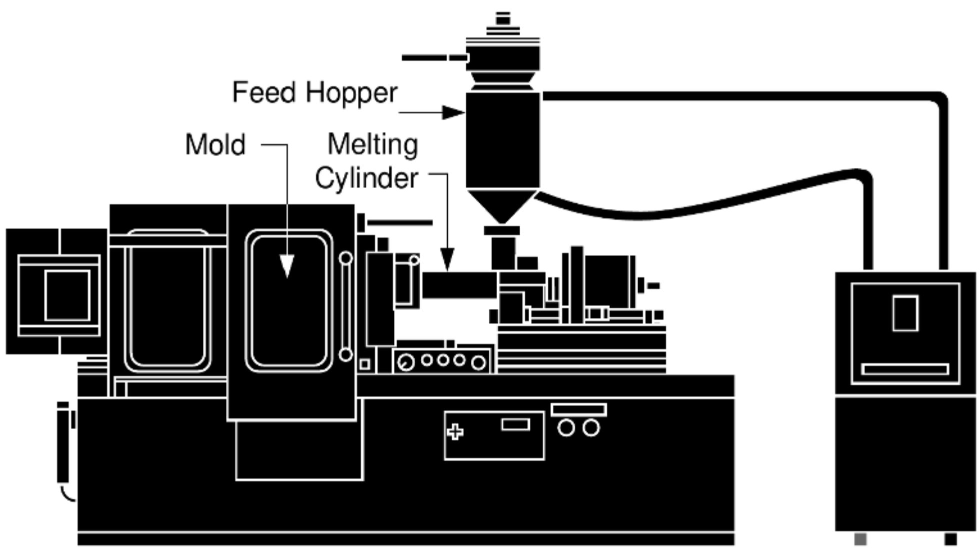

Figure 2.1: Injection molding machine schematic. Credit: (DuPont Engineering Polymers, 2000)

2.2 Manufacturing Domain

2.2.1 Core Principles of DFM for Injection Molding

When designing for injection molding, it is crucial to consider multiple factors that directly affect the manufacturability, quality, and cost of the final part. Adhering to established DFM principles mitigates production risks, reduces tooling complexity, and enhances the product's structural and cosmetic integrity.

2.2.2 Draft Angles

A draft angle is a slight taper applied to the faces of the part that are parallel to the mold's direction of pull (Oberg & Jones, 2020). This is a fundamental requirement to ensure that the part can be easily and cleanly ejected from the mold. Without an adequate draft, the friction between the part's surface and the mold wall is significant, which can lead to several manufacturing defects. There is a high risk of cosmetic damage, such as drag marks or scratches on the part's surfaces. Furthermore, the high forces required for ejection can introduce stress, leading to warping or even fracturing of the part (Oberg & Jones, 2020).

Figure 2.2 shows a diagram of an exaggerated draft angle. Notice the draft has been applied to both the inner and outer faces. While a general guideline is to apply a minimum draft of 1 to 2 degrees (Xometry Team, 2025), the optimal angle depends on factors like wall depth, material choice, and surface texture. Textured surfaces, for instance, require a greater draft to prevent the finish from being scraped off during ejection (Oberg & Jones, 2020). The complexity of manually inspecting every surface on a complex geometric part makes this a time-consuming and error-prone task, highlighting the need for an automated analysis tool that can instantly identify non-compliant faces.

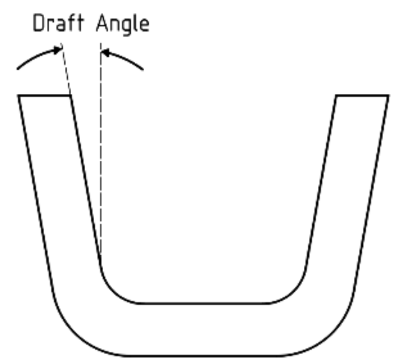

Figure 2.2: Draft angle diagram

Table 2.1: General guidelines for minimum draft angles in injection molding. (Xometry Team, 2025)

Application / Surface Description	**Minimum Draft Angle**
For "near-vertical" requirements	0.5°
General applications	2°
All shutoff surfaces	3°
Lightly textured faces	3°
Medium to heavily textured faces	5°+

Table 2.2: General Draft Angle Guidelines for Common Materials (HLH Rapid, n.d.-a)

Material	**Minimum Draft (°)**	**Recommended Draft (°)**
Nylon (PA)	0.0	1.0
Polyethylene (PE)	0.5	1.5
Polyvinyl Chloride (PVC)	0.5	1.5
Polypropylene (PP)	1.0	2.0
Polycarbonate (PC)	1.5	2.0

2.2.3 Wall Thickness

Maintaining a uniform wall thickness is one of the most critical principles in designing for injection molding (Oberg & Jones, 2020). Molten plastic is engineered to flow through the mold cavity and fill the part's geometry, but variations in thickness disrupt this process. Thicker sections cool much more slowly than thinner sections. This differential cooling causes uneven shrinkage (DuPont Engineering Polymers, 2000), which induces internal stresses in the part. These stresses can manifest as significant quality defects, including sink marks (small depressions on the surface opposite a thick feature), voids (internal bubbles), and warping (Maier, 2021).

The primary design goal is therefore to create a part with a consistent nominal wall thickness to promote uniform flow and cooling. Recommended wall thicknesses for various plastics can be found in Table 2.3. Large, solid features should be "cored out" (Maier, 2021) and replaced with a network of thinner structural ribs to provide the necessary support without creating thick masses of material. If variations in thickness are absolutely unavoidable, the transition between sections should be gradual and smooth rather than abrupt (Maier, 2021). See Figure 2.4.

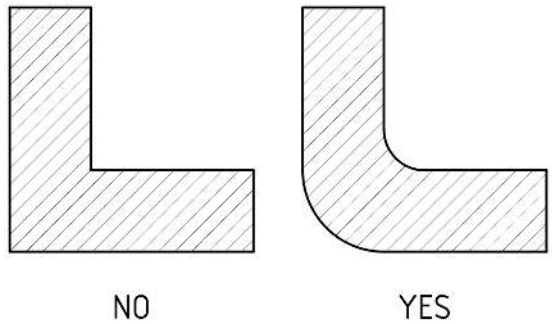

Figure 2.3: Uniform corner thickness. Inspired by DuPont Engineering Polymers (2000).

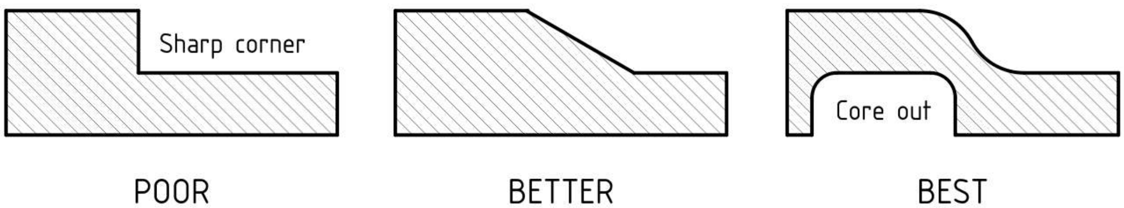

Figure 2.4: Thickness transition guide. Inspired by DuPont Engineering Polymers (2000).

Table 2.3: Recommended wall thickness for common injection molding materials (Xometry Team, 2025)

Material	**Recommended wall thickness (mm)**	**Recommended wall thickness (in)**
ABS	1.14 – 3.56 mm	.045 – .140 in
Acetal	0.76 – 3.05 mm	.030 – .120 in
Acrylic (PMMA)	0.64 – 12.70 mm	.025 – .500 in
Liquid Crystal Polymer	0.76 – 3.05 mm	.030 – .120 in
Long-Fiber Reinforced Plastics	1.91 – 27.94 mm	.075 – 1.100 in
Nylon	0.76 – 2.92 mm	.030 – .115 in
PC (Polycarbonate)	1.02 – 3.81 mm	.040 – .150 in
Polyester	0.64 – 3.18 mm	.025 – .125 in
Polyethylene (PE)	0.76 – 5.08 mm	.030 – .200 in
Polyphenylene Sulfide (PSU)	0.51 – 4.57 mm	.020 – .180 in
Polypropylene (PP)	0.89 – 3.81 mm	.035 – .150 in
Polystyrene (PS)	0.89 – 3.81 mm	.035 – .150 in
Polyurethane	2.03 – 19.05 mm	.080 – .750 in

Visualising wall thickness across an entire part is non-trivial, and identifying areas of unwanted material concentration or thinness is a key challenge that software analysis is uniquely equipped to address.

2.2.4 Undercuts

An undercut is a feature in a part design, such as a side hole or a snap-fit latch, that would prevent its direct ejection from a simple two-part mold. Because these features are not aligned with the pull direction, they would lock the part within the mold cavity. See Figure 2.5 for a depiction of an undercut. Undercuts can sometimes be worked around by moving them to be on the parting line (Proto Labs, Inc., 2021).

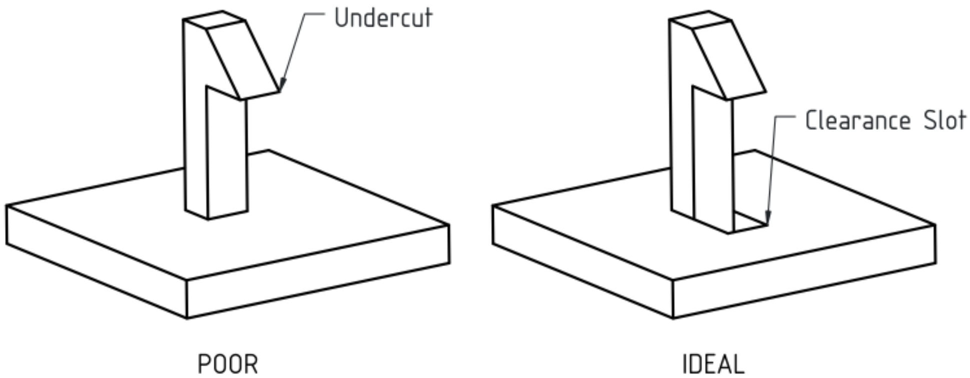

Figure 2.5: Depiction of a common undercut

Manufacturing parts with undercuts requires significantly more complex and expensive tooling (Xometry Team, 2025). These molds must incorporate moving components known as side-actions or lifters. These mechanisms slide into place to form the undercut feature and must then retract before the main mold opens for part ejection. The addition of these mechanisms dramatically increases the initial cost and lead time for mold fabrication, adds complexity to its ongoing maintenance, and can increase the production cycle time for each part (Xometry Team, 2025). As a result, a key DFM objective is to avoid or eliminate undercuts wherever possible to reduce overall manufacturing cost and complexity. An automated DFM tool would provide much insight to an engineer during the design phase, such that the undercuts could be eliminated before being sent off for manufacturing review by the supplier.

2.2.5 Corner Radii

Sharp internal corners in a part's design are a source of multiple issues in injection molding. From a flow perspective, it is difficult for molten plastic to navigate and properly fill sharp, 90-degree corners (BASF, n.d.). More critically, sharp internal corners create a high stress concentration (BASF, n.d.). This concentration of stress acts as a weak point, significantly compromising the part's mechanical strength and making it susceptible to failure when placed under load.

To mitigate these risks, all corners should be designed with a generous radius. Applying fillets to all internal and external corners distributes stress more evenly, greatly improving the part's durability (Maier, 2021). A generous radius also improves the plastic flow within the mold. Furthermore, rounded corners are easier and less costly to machine into the mold cavity, as rotating toolbits naturally create filleted features. To maintain uniform wall thickness (t), an internal radius of $0.5t$ and an external radius of $1.5t$ is advised (BASF, n.d.).

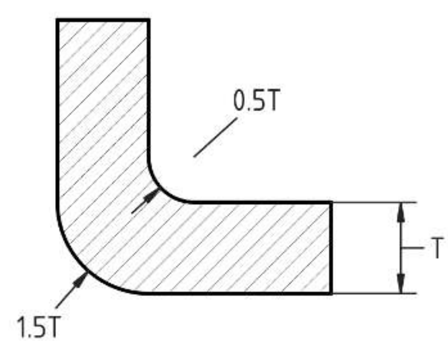

Figure 2.6: Recommended corner radii

2.2.6 Ribs

Ribs are a reinforcing feature that improve a part's structural rigidity. They should only be used when crucial for the parts performance or to improve mold flow to another feature. Excessive use of ribbing in a part can cause warping of the plastic and points of high stress concentration.

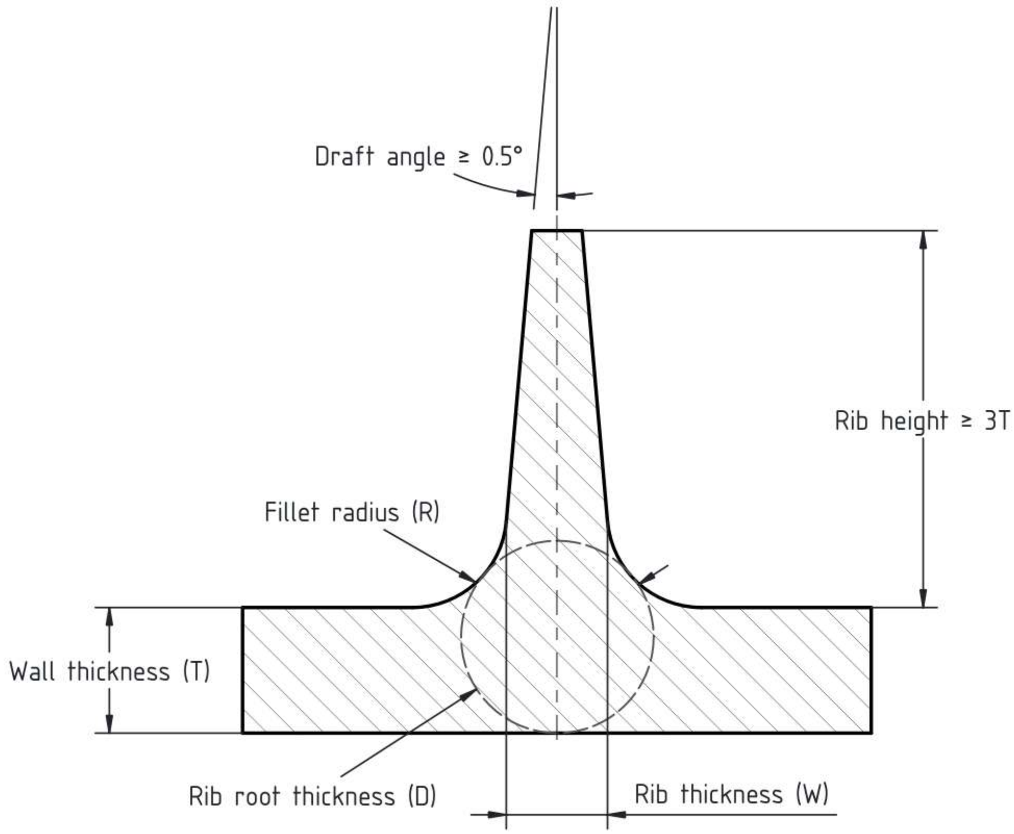

Figure 2.7: Rib design guide. Inspired by Maier (2021).

Table 2.4: Summary of DFM Guidelines for Rib Design in Injection Moulding (HLH Rapid, n.d.-c)

Design Parameter	**Recommended Value**	**Justification**
Rib Height	$\leq$ 3x Nominal Wall Thickness	Prevents incomplete filling (short shots) and breakage during part ejection.
Rib Thickness	40% – 60% of Nominal Wall Thickness	Critical for preventing cosmetic sink marks on the opposite surface due to uneven cooling.
Base Radius	0.5x – 1x Nominal Wall Thickness	Reduces high stress concentration at the base and improves molten plastic flow.
Rib Spacing	$\geq$ 2x Nominal Wall Thickness	Ensures the mould steel between ribs is strong enough and promotes uniform cooling.
Draft Angle	$\geq 0.5°$ per side	Facilitates a clean release of the part from the mould without drag marks.

2.2.7 Bosses

Bosses are often used as a mounting feature for external parts or as reinforcement around a hole. Table 2.5 details boss design best practices.

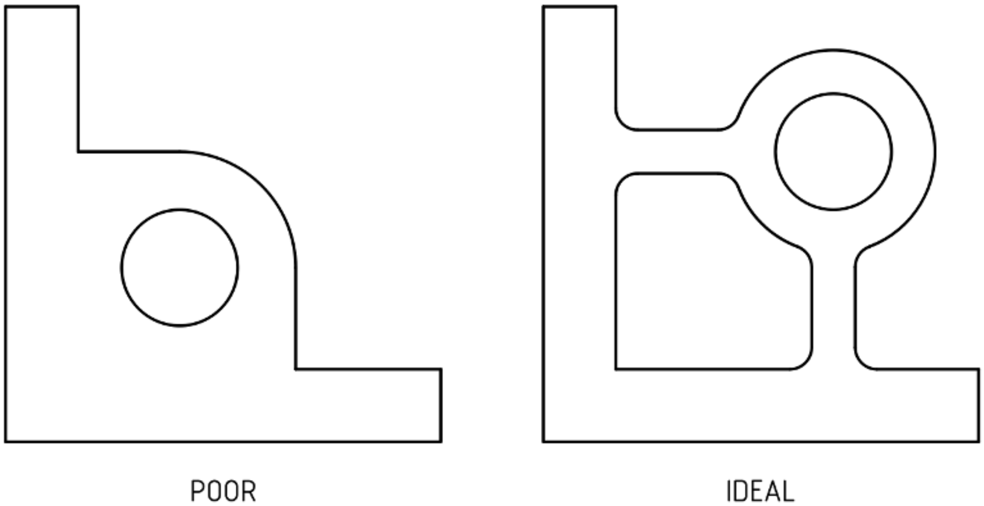

Figure 2.8: Boss design guide. Inspired by DuPont Engineering Polymers (2000)

Table 2.5: Summary of DFM Guidelines for Plastic Boss Design (HLH Rapid, n.d.-b)

Design Parameter	**Recommended Value**	**Justification**
Boss Placement	Distribute placement. Close to thicker walls. On interior of part.	Avoids concentrating stress and heat in one area, reducing risk of warp and defects.
Wall Thickness	$\leq$ 60% of Nominal Wall	Prevents cosmetic sink marks and internal voids caused by excess material.
Support Features	Use gussets, ribs, or connect to a main wall	For structual support.
Height-to-Diameter Ratio	Height < 3x Outer Diameter	Limits mechanical stress on the boss and ensures it can be filled properly during moulding.
Thread Engagement	Length $\geq$ 60% of Screw Thread	Ensures sufficient surface area for the fastener to grip securely.
Boss Spacing	$\geq$ 2x Nominal Wall Thickness	Maintains the integrity of the mould steel between features and promotes uniform cooling.
Draft (Inner Diameter)	$\geq 0.25°$	Allows the core pin to be removed easily without damaging the inner surface.
Draft (Outer Diameter)	$\geq 0.5°$	Facilitates a clean part release from the mould cavity without drag marks.
Base Radius	0.25x – 0.5x Nominal Wall	Critically reduces stress concentration at the base, preventing the boss from breaking under load.

2.2.8 Material

When manufacturing any plastic part, it is important to consider the material the part will be made from. Table 2.6 summarises common materials used in injection moulding, their properties and purposes.

Table 2.6: Summary of Common Materials for Injection Moulding (HLH Rapid, n.d.-d)

Material (Acronym)	Key Properties	Common Applications
PMMA (Acrylic)	Transparency, High Gloss, UV Resistance	Lenses, Lighting, Transparent Parts
PC (Polycarbonate)	Impact Resistance, High Strength, Clarity	Safety Equipment, Machinery Guards
ABS	General Purpose, Low Cost, Good Aesthetics	Consumer Products, Housings, Automotive
PP (Polypropylene)	Low Cost, Flexibility, Chemical Resistance	Living Hinges, Containers, Power Tools
PA (Nylon)	High Strength, Durability, High Temperature	Gears, Screws, Mechanical Parts, Snap-fits
PEEK	High Performance, High Cost, Extreme Temperature	Aerospace, Bio-medical, Electrical
POM (Acetal)	High Strength, Low Friction, Wear Resistance	Gears, Bearings, Bushings, Moving Parts
TPE	Rubber-like, Flexibility, Overmoulding	Soft-touch Grips, Seals, Flexible Parts
TPU	Hard Rubber, Elasticity, Abrasion Resistance	Gaskets, Wearables, Medical Tubing
Common Material Blends		
PC/ABS	High Strength, Heat Resistance, Processability	Automotive, Electronic Housings
PC/PBT	High Strength-to-Weight, Toughness, Chemical Resistance	Industrial Equipment, Electronics Covers

3 | Synthesis and Problem Statement

The literature clearly states a need for automation of checking designs against DFM principles. Specifically, injection moulding has presented itself as the perfect candidate for such automation against existing standards. Table 3.1 lists each DFM principle that should be checked in an automated DFM tool for injection moulding.

Table 3.1: Key DFM Checks and Justifications Identified from Literature Review

Automated Check	Justification / Core Purpose
Draft Angle	Ensures clean part ejection from the mould, preventing cosmetic defects (drag marks) and mechanical stress.
Wall Thickness	Promotes uniform cooling and plastic flow to prevent sink marks, voids, and warping.
Undercuts	Identifies features that would require complex and costly side-action tooling, allowing for cost-saving redesigns.
Corner Radii	Improves the flow of the molten plastic and mitigates stress concentration at internal corners to improve the part's mechanical strength and durability.
Rib Design	Validates that thin, structural ribs are used correctly to add rigidity without causing cosmetic defects like sink marks.
Boss Design	Validates that mounting and alignment bosses are designed to be strong while preventing sink marks and internal voids.

3.0.1 Research Question

A research question was formulated to guide the research and development of an automated DFM workbench for FreeCAD.

> *"Can a modular and extensible design for manufacturing workbench be created for FreeCAD, such that a design can be run against a set of defined rules and generate iterative useful feedback."*

4 | Methodology

4.1 Research Approach: Design Science

The design project will be developed in according to the Design Science Research (DSR) methodology (Hevner et al., 2004), a well established paradigm for research in information systems and engineering. The DSR method is well suited to this project due to its focus on deliverable artefacts, such as algorithmic specifications and UX competitor analysis.

A comprehensive roadmap was developed to guide the delivery of the DFM Workbench for FreeCAD, succinctly outlined in A.1 with proposed timelines. Each phase listed in A.1 will be comprehensively outlined in this chapter.

4.2 Research Phases

4.2.1 Phase 1: Research and Proof-of-Concept

This foundational phase is dedicated to rigorous problem definition and technical de-risking, aligning with the initial stages of the Design Science Research (DSR) cycle. The primary objective is to translate the identified practical need into a set of formal requirements and to verify the core technical feasibility of the proposed solution before committing to a specific software architecture. This phase ensures that the project is both relevant to its stakeholders and achievable within the chosen technical environment.

The initial activities are focused on requirement elicitation and specification, culminating in the creation of two key research artefacts:

- **Stakeholder Needs Analysis:** The research begins by engaging with the target end-users and developers within the FreeCAD community. This process validates the problem statement and ensures that the project's goals are directly aligned with real-world user needs.
- **System Requirements Specification:** Synthesising the findings from the literature review, this document formally defines the essential DFM rules and checks (e.g., minimum draft angle, maximum wall thickness) that the workbench must be able to perform. This artefact establishes the functional scope of the project.
- **Market Competitor Specification:** An analysis of existing commercial DFM tools (e.g., SOLIDWORKS DFMXpress, DFMPro) will be conducted. This involves evaluating their functionality, user workflows, and methods of presenting feedback. This comparative analysis establishes a baseline for usability and feature parity, informing the design of a professional-grade user experience.

The second component of this phase is a critical proof-of-concept (PoC) exercise designed to mitigate technical risk. A basic, command-line draft angle checker will be developed. This PoC deliberately avoids any complex architectural design choices, focusing purely on the core computational challenge: interacting with FreeCAD's underlying geometry kernel to perform a DFM calculation on a 3D model. The result—a simple pass/fail message printed to the terminal—serves two vital purposes: it provides early, definitive proof that the proposed analysis is technically feasible, and it allows for essential familiarisation with the geometric libraries. By successfully completing this phase, the project can proceed with the confidence that its core premise is sound.

4.2.2 Phase 2: Prototype Development

Following the successful de-risking and requirement specification of the initial phase, Phase 2 represents the first major design and build cycle of the DSR methodology. The central objective is to develop a functional prototype that implements the core analysis capabilities of the DFM workbench. This phase prioritises the implementation of the most critical and computationally distinct checks—draft angle and wall thickness—to serve as a testbed for both the analysis algorithms and the subsequent architectural design.

A key output of this phase is the *Algorithmic Implementation Specification*. This document will detail the underlying mathematical and computational logic required to perform the geometric analyses. It will precisely define the algorithms used for calculating draft angles on B-rep faces relative to a pull direction, as well as the methods for determining minimum, maximum, and uniform wall thickness across a solid body.

To ensure the rigour and objectivity of the evaluation process, a comprehensive **test suite of `.step` models** will be developed. This collection of 3D parts will be specifically designed with known geometric properties that intentionally pass or fail the DFM checks under specific conditions (e.g., a model with a 2.8-degree draft face to test a 3-degree rule). This suite enables repeatable, automated testing, forming the basis for validating the correctness of the implemented checks.

The development work will produce fully functioning analysers for draft angle and wall thickness. In a significant step beyond the command-line PoC, these checks will be integrated into the FreeCAD user interface. The user will be able to initiate an analysis from a basic setup Task Panel, with the textual results being presented in the standard FreeCAD Report Panel.

Crucially, the final deliverable of this phase will be the *Code Architecture Specification*. This document will be produced *after* the initial implementation of the checks. This deliberate sequencing embodies a reflective design practice; the process of building the first analysers will serve as a practical learning experience, exposing the challenges and requirements that a truly robust, scalable, and extensible architecture must address. Based on this experience, the specification will formally propose the design strategies used for this workbench and the high-level code flow. It will also define the definitive folder and file structure for the workbench, establishing the foundational design that will be fully implemented in subsequent phases.

4.2.3 Phase 3: Minimum Viable Product (MVP)

This phase embodies the Minimum Viable Product (MVP) development paradigm, where the focus shifts from implementing discrete features to delivering a complete, polished, and self-contained user experience. The primary objective is to create a version of the DFM Workbench that delivers the core value proposition of the project: a seamless workflow from model selection to the clear, interactive visualisation of design flaws. By establishing a stable, documented, and deliverable product at this stage, the project's core success is de-risked. All subsequent feature development can then proceed on separate branches from a known-good foundation, ensuring that a "shippable" version of the project is always maintained.

The central development effort will be the creation of a fully functional results task panel. This moves the user workflow entirely into the workbench's dedicated GUI, replacing the temporary use of the FreeCAD Report Panel. This holistic integration is critical for creating an intuitive and professional user journey, where all interactions occur within a single, consistent interface.

A significant technical challenge and the cornerstone feature of this phase is the implementation of interactive 3D highlighting. Upon selecting a result in the task panel, the workbench will highlight the corresponding problematic geometry (faces, edges) directly in the 3D view. This requires direct interaction with FreeCAD's underlying 3D scene graph (Coin3D) and represents the most critical step in providing actionable, iterative feedback to the designer. This feature transforms the workbench from a simple reporting tool into an interactive guidance system.

Finally, to ensure the MVP is a truly complete and deliverable package, comprehensive user and developer documentation will be created. The user documentation will guide designers on how to install and effectively use the workbench, while the developer documentation will provide an initial overview of the code architecture and explain how third parties could begin to extend the tool with new checks. Completing this documentation ensures that, if necessary, the project in its MVP state could be handed over and understood, fully mitigating risks to final project delivery.

4.2.4 Phase 4: v0.1 Release and Architectural Validation

This phase marks the transition from a minimum viable product to a feature-rich, robust, and extensible platform. The primary objective is to fully instantiate the architectural vision defined in the *Code Architecture Specification*, thereby validating its design through practical implementation and extensibility.

The development will commence with the implementation of the undercut check, a geometrically distinct analysis that serves as an excellent test case for the newly implemented plugin architecture. Successfully integrating this check will provide the first concrete evidence that the system is genuinely modular. Following this, the architectural groundwork will be fully realised, building a system that is not only capable of handling all planned checks for injection moulding but is also explicitly designed for future extension. Hooks and base classes will be established to allow other developers to easily contribute analysers for entirely different manufacturing processes, such as FDM 3D Printing or CNC Machining.

A key deliverable for empowering users is the introduction of customisable standards. The system will be implemented to read from and potentially write to user-editable configuration files, allowing parameters such as the minimum draft angle or recommended wall thickness to be modified to suit specific materials or project requirements.

Critically, this phase focuses heavily on elevating the user experience from simple feedback to active learning. Local HTML help pages will be created and integrated directly into the results panel. When a user receives a finding, they will have immediate access to a concise, illustrated explanation of *why* the check is important, the manufacturing defects it prevents, and common strategies for remediation. This feature transforms the workbench into a pedagogical tool, which is vital for the target audience of hobbyists and pro-amateurs who may lack deep industry knowledge of DFM principles. To further refine the interactive workflow, auto-zooming to problem geometry will be implemented as a quality-of-life feature, immediately directing the user's attention to the relevant area of the model.

The culmination of this phase will be the first stable v0.1 release to the broader FreeCAD community. This release serves as a crucial evaluation step in the DSR cycle, intended to gather widespread user testing and qualitative feedback. A substantial buffer period will be allocated in the subsequent phase to analyse this feedback, address any reported bugs, and make informed decisions for the final stage of development.

4.2.5 Phase 5: Stretch Goals and Final Evaluation

This final phase of the research project is structured around a "stretch goal" philosophy, serving as a sophisticated de-risking strategy. The features planned for this stage are not promised as core deliverables but are identified as high-value extensions that will be pursued if the preceding phases are completed on schedule. By categorising these complex tasks as contingent objectives, the project ensures it can deliver a complete and successful core product, while still allocating time to explore advanced capabilities that would significantly enhance the workbench's power and utility.

A significant portion of this phase is allocated to the investigation of highly complex DFM checks for bosses, fillets, and ribs. These features represent a paradigm shift from the global geometric analysis of previous phases to localised, automatic feature recognition. The development of a robust feature recognition engine is a substantial research and implementation challenge in itself, which is why these checks are prudently delegated to this final, exploratory stage.

Further extensions to the core functionality will be explored, including support for secondary pull directions to handle more complex part geometries, and the ability to export a formal DFM report in a standard format (e.g., CSV spreadsheet). These features would move the workbench's capabilities closer to those of mature commercial products.

Crucially, this final evaluation cycle is also responsive in nature. Substantial time is reserved for systematically addressing bug fixes and incorporating user-requested features gathered from the community feedback on the v0.1 release. This iterative refinement is a vital part of the DSR process, ensuring the final artefact is not only functional but also robust and aligned with user expectations. The culmination of these efforts will be the final polish of the product in preparation for the project demonstration, ensuring a stable, well-documented, and impactful research artefact.

4.3 Ethical, Indigenous, and Sustainability Considerations

A core tenet of engineering practice in Australia is the consideration of a project's broader societal and environmental impact. This research, while technical in nature, is grounded in a framework of ethical conduct, acknowledges its context within Australia's reconciliation journey, and aims to contribute positively to sustainability goals.

4.3.1 Ethical Considerations

The ethical foundation of this project is built upon the principles of Free and Open-Source Software (FOSS). The choice to develop the workbench for FreeCAD, and to release the workbench itself under a FOSS license (LGPL-2.1), is a deliberate ethical decision. This approach ensures:

- **Transparency and Trust:** The source code will be publicly accessible, allowing for peer review of the algorithms and implementation. This transparency ensures the tool operates as described and builds trust within the user community, precluding any "black box" behaviour.
- **Democratisation of Technology:** By providing a professional-grade engineering tool at no cost, this project works to lower economic barriers to entry. This ensures that access to powerful design tools is not limited to well-funded corporations but is available to students, hobbyists, startups, and independent engineers across Australia and the world.
- **Data Privacy and Security:** The DFM workbench is designed to run entirely locally on the user's machine. It does not require an internet connection to function and does not collect or transmit any user data, designs, or results to an external server. This respects user privacy and protects their intellectual property by design.

4.3.2 Indigenous Dimensions

While this project does not involve direct engagement with First Nations communities, it has been designed with an awareness of the principles outlined in national initiatives such as the Closing the Gap agreement. The project's contribution is indirect but principled, focused on fostering a more equitable and accessible technological landscape.

The commitment to a free and open-source platform is paramount. The significant cost of commercial CAD software can be a major obstacle for community-controlled organisations, regional TAFEs, and aspiring Indigenous-owned businesses. By enhancing the capabilities of a free platform, this project contributes a resource that can support Indigenous participation in STEM. It provides a tool that can be used in educational programs, makerspaces, or by entrepreneurs to design and validate products locally, fostering technological sovereignty and supporting pathways to economic self-determination without requiring significant capital investment. This aligns with the broader goal of ensuring all Australians have the opportunity to participate in and benefit from the digital economy.

4.3.3 Sustainability Considerations

The project directly addresses sustainability by embedding principles of waste reduction into the earliest stages of the product design lifecycle. Design for Manufacturability is, by its nature, a practice that enhances material and energy efficiency. This workbench contributes to sustainability in the following ways:

- **Reduction of Material Waste:** The primary function of the tool is to identify design flaws before production tooling is made or manufacturing commences. For a process like injection moulding, this can prevent the creation of expensive, faulty moulds and the production of thousands of scrap parts. In the context of future extensions for 3D printing, it reduces the number of failed prints, directly saving polymer materials.
- **Conservation of Energy:** Manufacturing processes are energy-intensive. By minimising production errors and rejected parts, the workbench helps to reduce the associated energy consumption, contributing to a lower carbon footprint for the manufacturing process.
- **Improved Product Durability:** DFM principles, such as ensuring correct corner radii to mitigate stress concentrations, result in stronger, more durable parts. A more durable product has a longer service life, which aligns with the principles of a circular economy by reducing the need for premature replacement and subsequent waste.

5 Conclusion

This research proposal has outlined a structured and phased plan to develop a novel Design for Manufacturability (DFM) workbench for FreeCAD. The project directly addresses a critical capabilities gap in the open-source ecosystem by providing users with automated, interactive feedback to improve the quality and manufacturability of their designs.

Upon successful completion, this project will yield two primary contributions. The first is a practical and high-utility software artefact that directly empowers the FreeCAD community, helping to prevent costly manufacturing errors and reduce material waste. The second, and more significant from a research perspective, is a contribution to design knowledge. By pioneering and documenting a modular and extensible architectural framework for analysis tools, this project will provide a validated blueprint for future development.

This ensures the workbench is not a monolithic tool, but a lasting platform that the community can easily build upon, extending its capabilities to new manufacturing processes. In essence, this project delivers both a valuable tool and the knowledge of how to create similar tools, making a meaningful and lasting impact on the open-source engineering landscape.

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A | Project Plan

A.1 Roadmap

Table A.1: High-Level Project Roadmap and Deliverables

Phase	Name	Timeline	Key Deliverables
1	Proof-of-Concept & Research	Aug - Sep 2025	• Stakeholder needs analysis. • Artefact *System Requirements Specification* • Artefact *Market Competitor Specification* • Basic draft check. Prints to terminal.
2	Prototype	Oct 2025	• Artefact *Algorithmic Implementation Specification* • Test suite of .step models for objective testing of checks • Fully functioning draft and thickness checks • Can be run from setup task panel GUI. Results in terminal. • Artefact *Code Architecture Specification*
3	Minimum Viable Product (MVP)	Nov 2025	• Results task panel functioning • Highlight problem geometry in 3D view • Developer and user documentation
4	v0.1 Release	Dec 2025 - Jan 2026	• "Code Architecture Specification" verdict implemented fully • Undercut check • Custom DFM standards, editable in GUI, saved locally. • Local HTML help pages accessible from results panel • Auto-zooming to problem geometry in 3D view
5	Stretch Goals	Feb - June 2026	• Boss, fillet, rib checks • Secondary pull directions • Export report (spreadsheet in .odt format) • User requested features

A.2 Resources

A.3 Team

Refer to Table A.2 for a list of all contributing team members. A contribution can be defined as any of the following; feedback, design discussions, review, code creation, icon creation, testing.
FreeCAD has multiple working groups, each with their own specialisation. These are:

- Design Working Group (DWG)
- CAD Working Group (CWG)
- Code Quality Working Group (CQWG)

Table A.2: List of contributors

Contributer	Role
Ryan Kembrey	Capstone student
Anna Lidfors-Lindqvist	Supervisor
Benjamin Nauck	Co-Supervisor, DWG, CQWG
@Pierreporte	CWG, DWG
@Max	CWG, DWG
@Obelisk	DWG

A.4 Deliverables

Table A.3: Table of deliverables

Deliverable	Due Date	Description
Literature Review	Sep 2025	PDF report summarising the existing research and identification of a research question
Project Proposal	Oct 2025	PDF report outlining the proposed capstone project.
MVP	Nov 2025	
FreeCAD DFM Workbench	June 2026	Hosted on GitHub, installable via the Addon Manager
Code documentation	Feb 2026	In markdown format
User documentation	Feb 2026 TBD	In markdown format